Programmatically Representing Data

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Overview

Overview

- From Files to Objects
- Working with Data Frame metadata
- From Objects to Files

JSON File

Key-value pair structured data

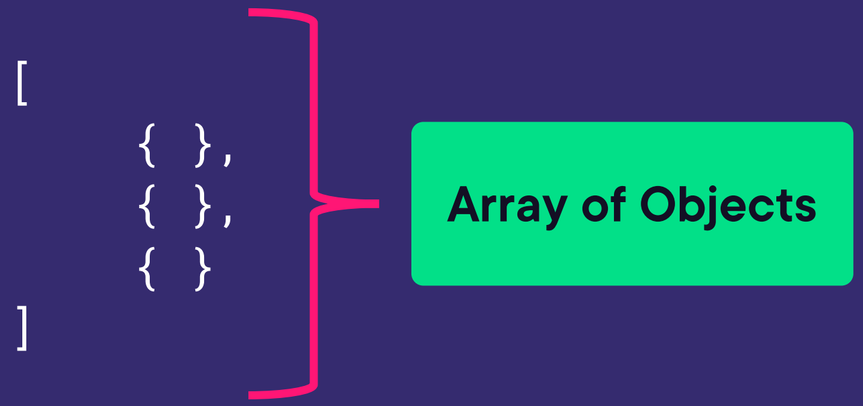
Example-1.json

```
{ "Key": "Value" }
```

JSON File

Key-value pair structured data

example-1.json



JSON File

Key-value pair structured data

example-1.json

Key: Column

Value: Cell Value

```
[
    { "Column 1": 1, "Column 2": 2 },
    { "Column 1": 3, "Column 2": 4 },
    { "Column 1": 5, "Column 2": 6 }
]
```

Array of Objects

From JSON file to Data Frame with `read_json()`

```
In [1]: import pandas as pd

In [2]: df_from_json = pd.read_json("example-1.json")

        df_from_json
```

Out[2]:

	Column 1	Column 2
0	1	2
1	3	4
2	5	6

Location of JSON file

```
In [3]: json_string = df_from_json.to_json()

        json_string

Out[3]: '{"Column 1":{"0":1,"1":3,"2":5},"Column 2":{"0":2,"1":4,"2":6}}'
```

From Data Frame to JSON String with `to_json()`

```
In [1]: import pandas as pd

In [2]: df_from_json = pd.read_json("example-1.json")

        df_from_json
```

Out[2]:

	Column 1	Column 2
0	1	2
1	3	4
2	5	6

```
In [3]: json_string = df_from_json.to_json()

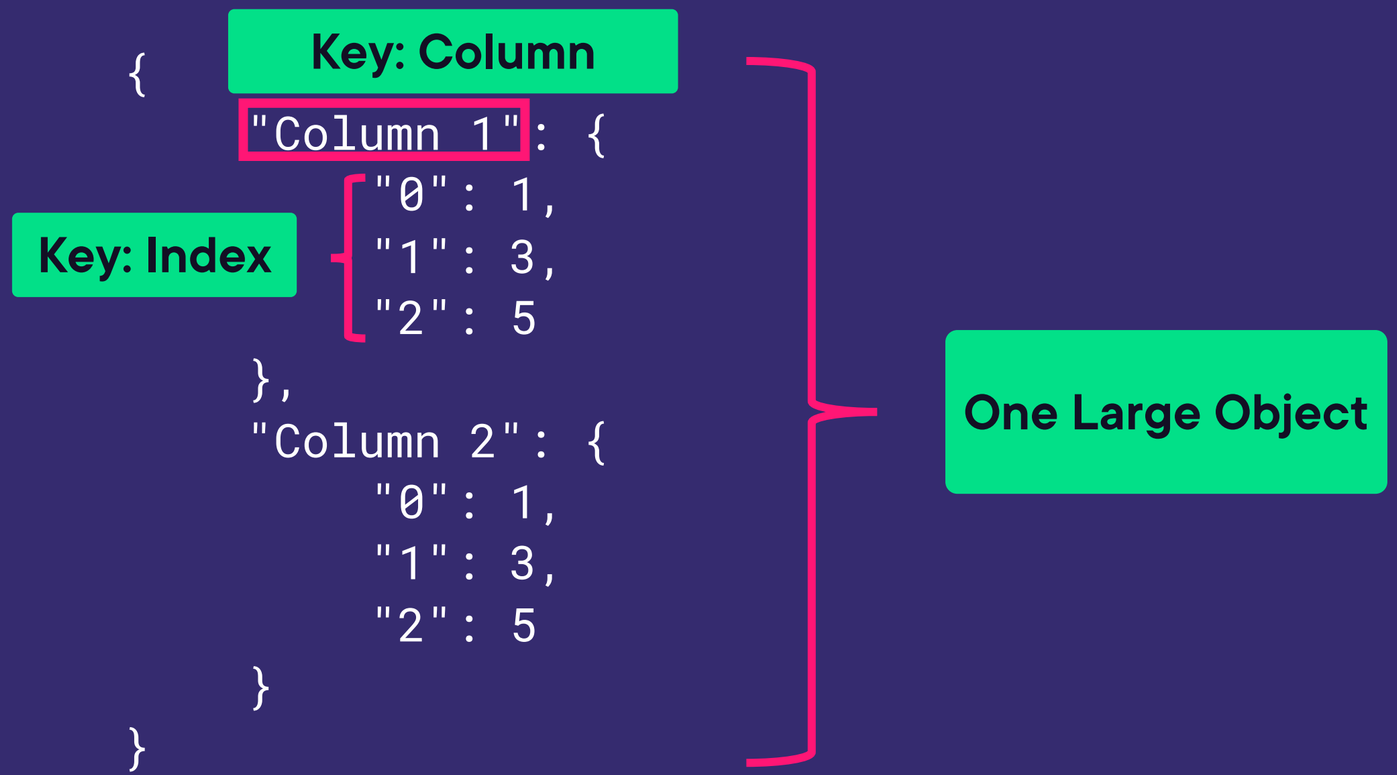
        json_string

Out[3]: '{"Column 1":{"0":1,"1":3,"2":5},"Column 2":{"0":2,"1":4,"2":6}}'
```

JSON File

Second example with a different structure

example-2.json




```
{
    "Column 1": {
        "0": 1,
        "1": 3,
        "2": 5
    },
    "Column 2": {
        "0": 1,
        "1": 3,
        "2": 5
    }
}
```

From JSON File to Data Frame

In [1]:
```
import pandas as pd
```

In [2]:
```
df_from_json = pd.read_json("example-2.json")

df_from_json
```

Out[2]:

	Column 1	Column 2
0	1	2
1	3	4
2	5	6

In [3]:
```
json_string = df_from_json.to_json()

json_string
```

Out[3]:
```
'{"Column 1":{"0":1,"1":3,"2":5},"Column 2":{"0":2,"1":4,"2":6}}'
```

CSV Example 1

Comma delimited text file with headers as first row

example-1.csv

First Row as Headers

```
Branch,Date,Amount
Branch A,January 1,500.00
Branch B,January 2,250.00
Branch C,January 3,300.00
```

From CSV file to Data Frame

```
In [1]: import pandas as pd

In [2]: df_csv = pd.read_csv("example-1.csv")

        df_csv
```

Location of CSV file

Out[2]:

	Branch	Date	Amount
0	Branch A	January 1	500.0
1	Branch B	January 2	250.0
2	Branch A	January 3	300.0

CSV Example 2

Comma delimited text file with first row as immediate data (no header info)

example-2.csv

First Row as Headers

```
Branch A,January 1,500.00
Branch B,January 2,250.00
Branch C,January 3,300.00
```

From CSV file to Data Frame (Incorrect)

```
In [1]: import pandas as pd

In [2]: df_csv = pd.rea_csv("example-2.csv")

        df_csv
```

Out[2]:

	Branch A	January 1	500.00
0	Branch B	January 2	250.0
1	Branch A	January 3	300.0

From CSV file to Data Frame (Correct)

```
In [1]: import pandas as pd

In [2]: df_csv = pd.read_csv("example-2.csv", header=None)

        df_csv
```

Out[2]:

	0	1	2
0	Branch A	January 1	500.0
1	Branch B	January 2	250.0
2	Branch A	January 3	300.0

Data Frame to CSV with `.to_csv()`

Saving to a file programmatically from a data frame object WITH index

script.py

```
df_data.to_csv("data.csv")
```

Location of CSV file

Indexes

data.csv

```
,0,1,2
0,Branch A,January 1,500.00
2,Branch B,January 2,250.00
3,Branch A,January 3,300.00
```

Data Frame to CSV with `.to_csv()`

Saving to a file programmatically from a data frame object WITHOUT index

script.py

```
df_data.to_csv("data.csv",index=None)
```

data.csv

```
Branch A,January 1,500.00
Branch B,January 2,250.00
Branch A,January 3,300.00
```

Demo

Working with External Files

- Reading from JSON
- Reading from CSV
- Saving to File

Summary

Summary

- Turn CSV files to Data Frames
- Basic operations and gotchas
- Turn Data Frames back to CSV

Up Next:

Exploring and Evaluating Data

Author Course Template

2022.10.b

View in Slide Show mode the first time you run through this deck.

Slides with the light bulb icon are guidelines only.
Please do not use them in your course.

Hello!
We've packed this slide template with options you can use to tell compelling stories to engage learners.

You'll notice that most of the slides have a white background, but some (like this one) have a black background. You'll find tips, tricks, and insights on the black slides, and you can use the white slides as templates or inspiration for your own content slides.

If you have questions, feedback, or suggestions for the deck, share them with your video content producer or post them in #ps_production-updates in Author Slack.

Happy authoring!
Content Production Services Design Team

More Resources

- Refer to the Creating Slides page for more guidance on using this deck.

- Use our icon library as your first source for images. Either open the searchable Icon Library page on Author Kit or download the latest version from the Downloads page.

- Use the Title Case Converter to meet our standards for capitalizing course names, module titles, clip titles, and slide titles.

Committing to Pluralsight Standards

Key Standards:

- Use only PS TT Commons font for content slides and Roboto Mono font for code or terminal output slides.
- Apply title case to all course, module, clip, and slide titles.
- Use only the Pluralsight color palette.
- Ensure all images are high-quality and copyright-safe.
- Use animations and slide transitions only if they call out important information or help tell a more compelling story.

More information: Slide Standards and Guidelines page

Pluralsight Watermark

All Pluralsight video content is required to have a Pluralsight watermark. This watermark can be seen in the lower right-hand corner (if you are in editor mode, you can see it in the lower right-hand corner of this slide).

The watermark is applied to your final video files when they are run through the encoding processes. This ensures that the mark appears over your slides as well as any demo footage in your course.

The mark you see here in PowerPoint is only for reference in the editor mode so that you do not put any content there. When viewing your slides in presentation mode (and ultimately recording them), the mark will not appear.

Accessibility

It’s important to make sure our slides are accessible to as many people as possible.

Throughout the slide deck, look for this symbol in the slide margins that indicates accessibility notes.

See what they look like right below this slide

Pluralsight Color Palette: Primary Colors

Transform Pink is our main primary color and should be more prominent than any of the secondary colors. Inky Blue is our other primary color and is used as our dark mode background and instead of black for body copy.

Transform Pink

Hex #FF1675

RGB 255,22,117

Inky Blue

Hex #130F25

RGB 19,15,37

Pluralsight Color Palette: Secondary Colors

Secondary colors can be used as accent colors for emphasis.

Hex #00A3FF RGB 0,163,255	Hex #770EF7 RGB 119,14,247	Hex #FFC942 RGB 255,201,66	Hex #FF7B01 RGB 255,123,1	Hex #02E088 RGB 2,224,136

Pluralsight Color Palette: Gray Alternative

These colors can be used as an alternative to gray or as lighter versions of Future Blue.

Hex #1B1834
RGB 27,24,52

Hex #585FA2
RGB 88,95,162

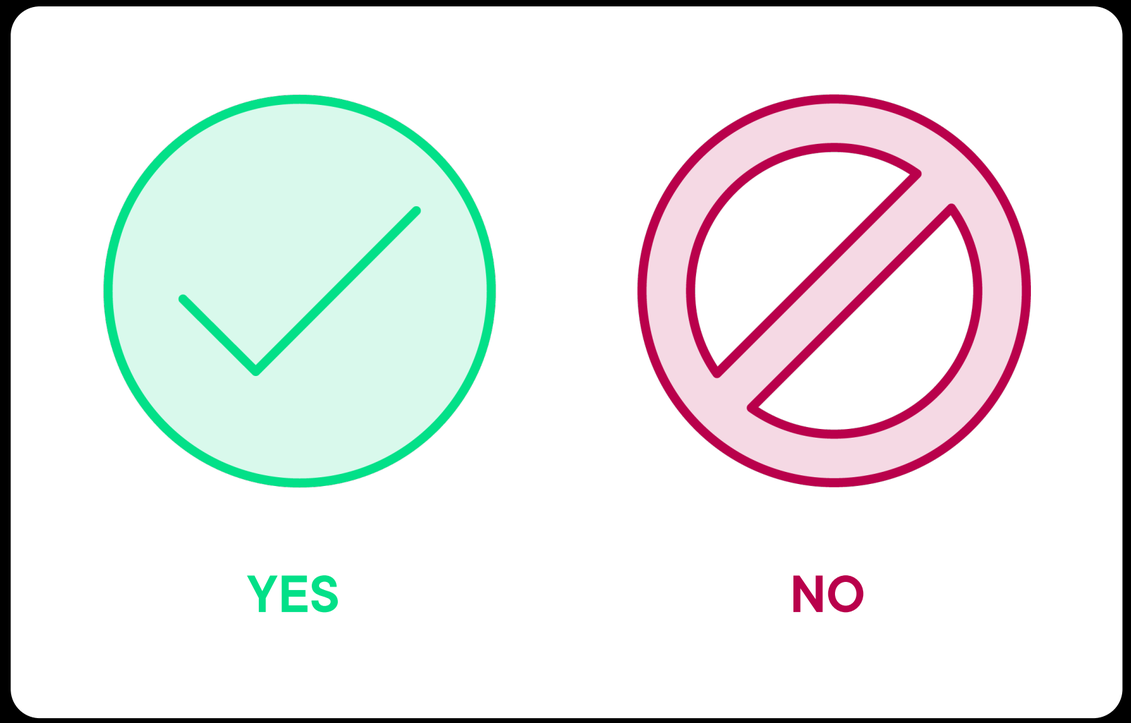
Hex #EBEFF5
RGB 235,239,245

Pluralsight Color Palette: YES vs. NO

Should the situation arise where you need to include YES vs. NO (go/stop) imagery, use our green and darkest shade of primary pink.

Hex #02E088
RGB 2,224,136

Hex #B9004B
RGB 185,0,75



Pluralsight Color Palette: Caution

Should the situation arise where you need to represent caution or a warning, use the yellow from our secondary color palette.

Hex #FFC942
RGB 255,201,66

Rounded Corners

All rectangles should have rounded corners.

When selecting a shape, choose the “rounded rectangle”.

By default, the edges will become more rounded as the shape gets larger. Decrease the corner radius by clicking and dragging the small yellow square that appears when you click on the rectangle.

Make sure the corner radius is consistent per slide. It should not be too large.

Correct

Incorrect

PS TT Commons & Roboto Mono Fonts

If these two columns match, PS TT Commons and Roboto Mono are correctly installed.

PS TT Commons Light

PS TT Commons Regular

PS TT Commons Demibold

Roboto Mono

PS TT Commons Light

PS TT Commons Regular

PS TT Commons Demibold

Roboto Mono

Font Sizes

As mentioned, the main font you will be using is PS TT Commons. Use Robot Mono for code or terminal output slides.

- **Body copy:** for maximum legibility, do not use less than a 24-point font size – 34-36 is standard, using PS TT Commons Demi Bold.
- You may use a font size less than 24 pt on diagrams when needed as long as legibility is not lost. It is not recommended to go below 18 pt.
- **Slide title:** use PS TT Commons Bold at 48-point font size. The title can be more than one line if needed.

Table of Contents

Template Types

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- Text Chunking
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- Bullet Lists
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- Diagrams
- Slide Transitions
- Other Slides

Title Slides

Using Title Slides

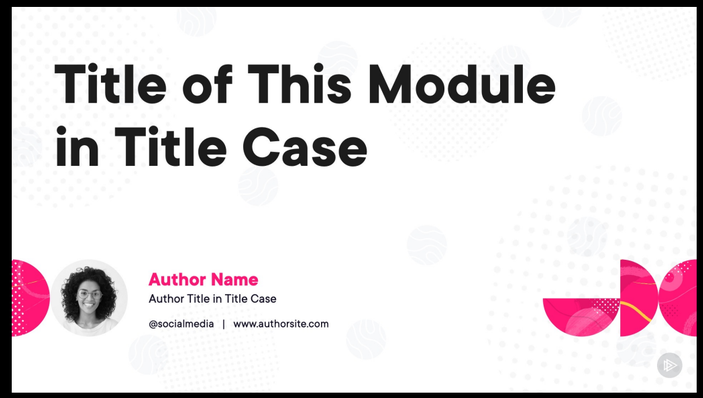
These templated title slides are required at the beginning of every module. Please do not alter or use any of these slides for anything other than their intended purposes.

Module TWO only *(your first content module):*

- Place the course title in the main space.
- Place the module two title underneath.

Module THREE and higher:

- Place the module title in the main space.



Using Section Header Slides

Section slides are optional. Please do not alter or use any of these slides for anything other than their intended purposes.

SECTION HEADER slide:

– Use this slide only at the beginning of a clip (optional - not required).

UP NEXT slide:

– Use this slide only at the end of a module (optional - not required).

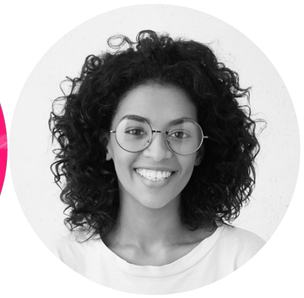
Replacing an Image

How to replace your image:

- Right-click the image you wish to replace.
- Right-click > Change Picture > From a File
- To create a black and white image, lower the color saturation to 0%.
- Color saturation can be found in the Picture Format Pane, under the Picture tab, then Picture Color.

Title of This Course in Title Case

Module Two Title in Title Case

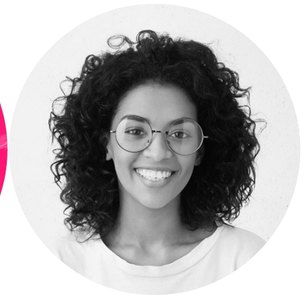


Author Name

Author Title in Title Case

@socialmedia | www.authorsite.com

Title of This Module in Title Case



Author Name

Author Title in Title Case

@socialmedia | www.authorsite.com

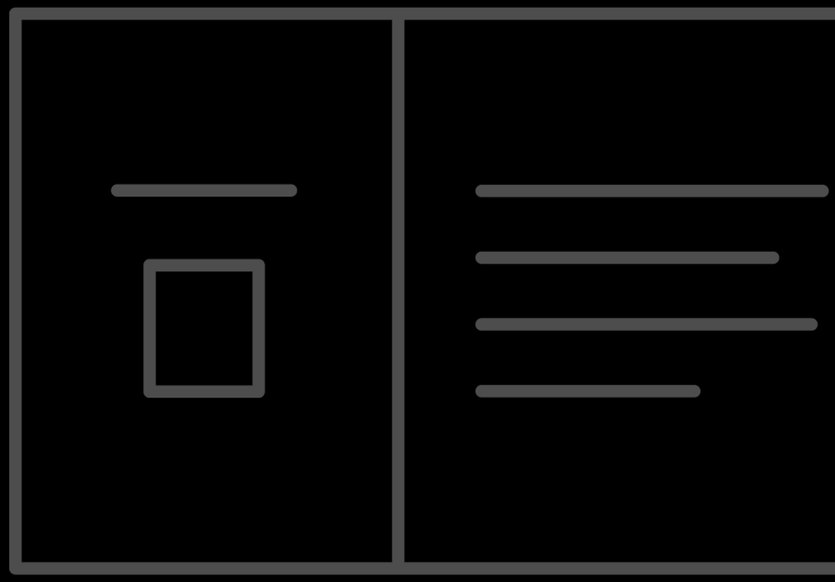
Add Section Header in Title Case

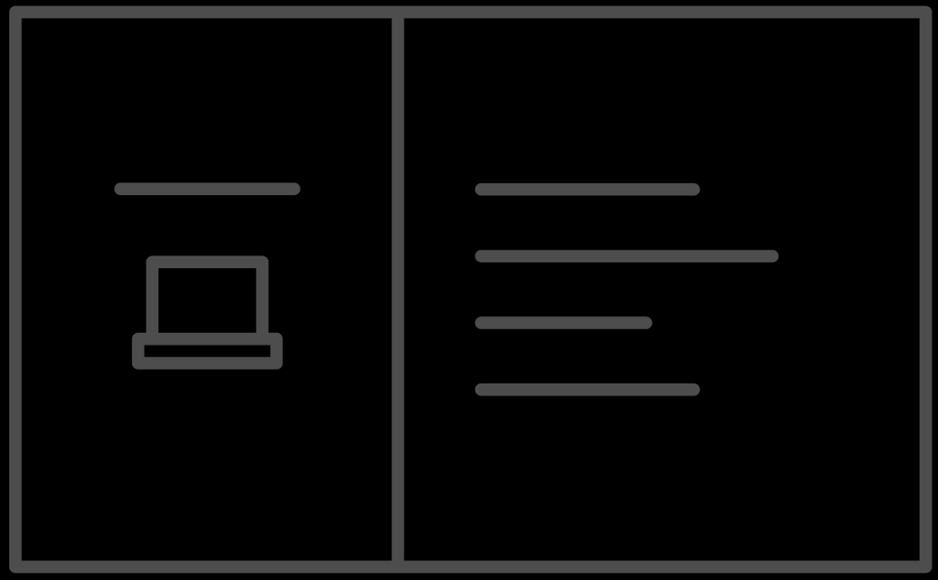
Up Next:

Title of Upcoming Module

TOC

Overview/Summary and Demo Layouts





Overview/ Summary

Use this layout to introduce or summarize

Overview

- Explain what's in this section for learners
- Layout a problem that will be solved

Summary

- Revisit problem and solution
- Provide practical takeaways

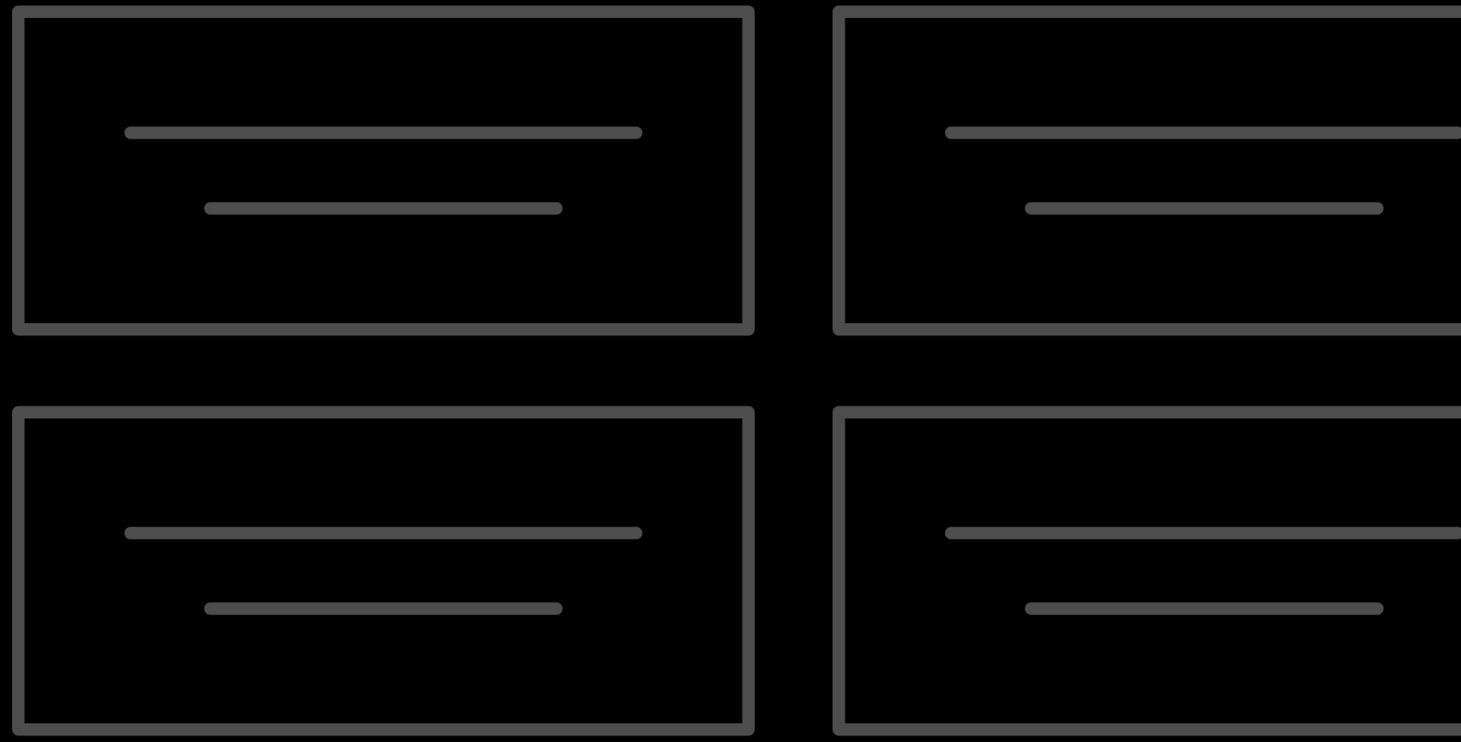
Demo

Use this layout to introduce your demo

Set the stage for this step

- Why we do it
- How we do it

Text Chunking



Text Chunking: Two Items with Headers

Talking point one

Be concise and keep the text to four lines or fewer

Talking point two

Be concise and keep the text to four lines or fewer

Text Chunking: Two Items

Be concise and keep the text to four lines or fewer

Be concise and keep the text to four lines or fewer

Text Chunking: Two Items Outlined

Be concise and keep the text to four lines or fewer

Be concise and keep the text to four lines or fewer

Text Chunking: Three Items

This is the first talking point

This is the second talking point

This is the third talking point

Text Chunking: Three Items Outlined

This is the first talking point

This is the second talking point

This is the third talking point

Text Chunking: Four Items

This is the first talking point

This is the second talking point

This is the third talking point

This is the fourth talking point

Text Chunking: Four Items Outlined

This is the first talking point

This is the second talking point

This is the third talking point

This is the fourth talking point

Text Chunking: Five Items

This is the first talking point

This is the second talking point

This is the third talking point

This is the fourth talking point

This is the fifth talking point

Text Chunking: Five Items Outlined

This is the first talking point

This is the second talking point

This is the third talking point

This is the fourth talking point

This is the fifth talking point

Text Chunking: Six Items

Thing 1

Thing 2

Thing 3

Thing 4

Thing 5

Thing 6

Text Chunking: Six Items Outlined

Thing 1

Thing 2

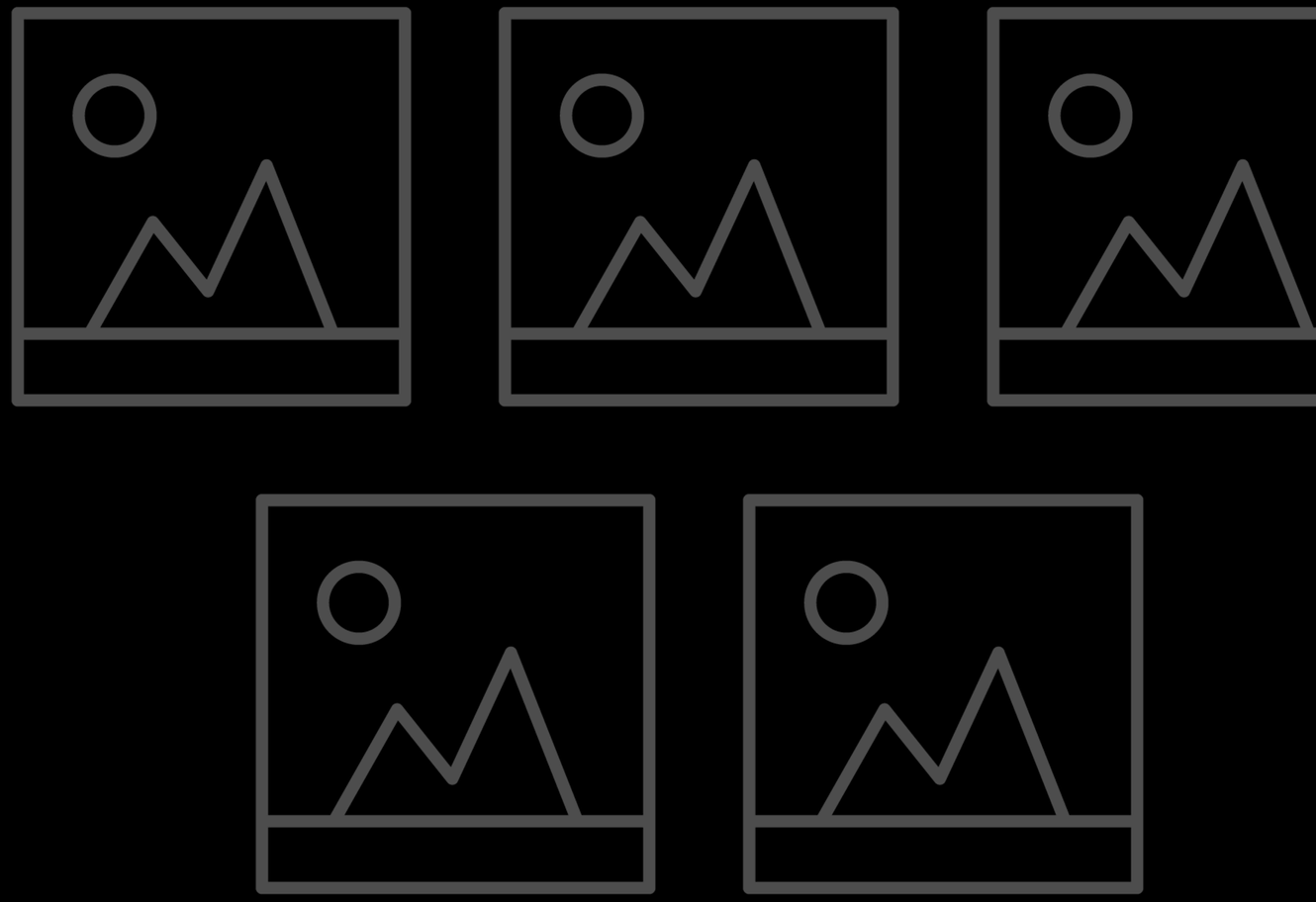
Thing 3

Thing 4

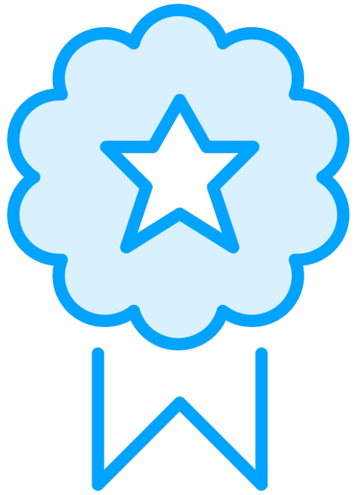
Thing 5

Thing 6

Image Chunking

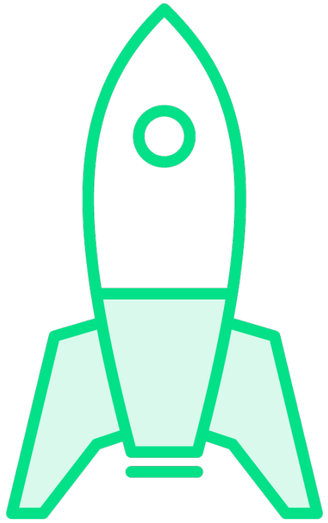


Icon or Image Chunking: Two Items with Headers



Informative Title

Some information about this graphic goes here and four lines or fewer is best



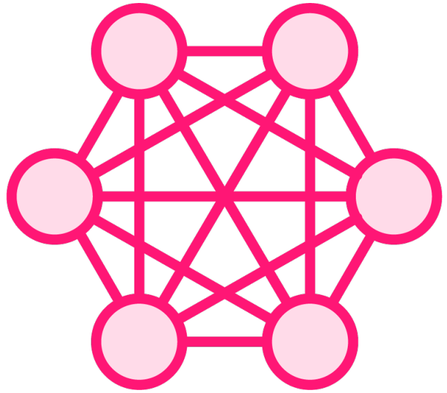
Informative Title

Some information about this graphic goes here and four lines or fewer is best

Icon or Image Chunking: Three Items

Some information about this graphic goes here

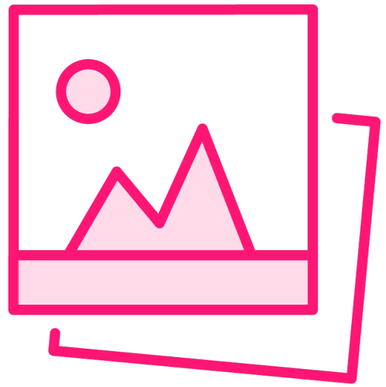
Some information about this graphic goes here



Some information about this graphic goes here

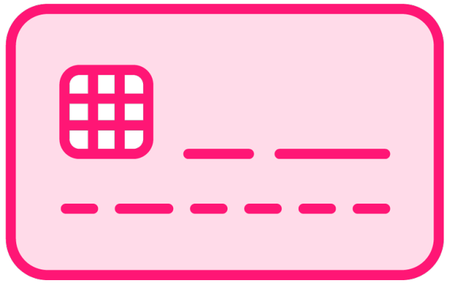
Icon or Image Chunking: Four Items

Some info goes here; three lines or fewer is best



Some info goes here; three lines or fewer is best

Some info goes here; three lines or fewer is best

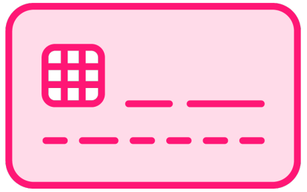


Some info goes here; three lines or fewer is best

Icon Only Image Chunking: Six Items

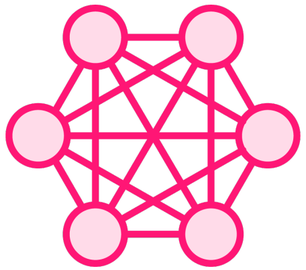
Data transfer

Email



ID card

Game controller



Network map

VR headset

Icon or Image Chunking: Two Items with Headers

Partner pairing

Some information about this graphic goes here and four lines or fewer is best

Brainstorming ideas

Some information about this graphic goes here and four lines or fewer is best

Circle Icon or Image Chunking: Two Items

Partner pairing

Brainstorming ideas

Icon or Image Chunking: Three Items

Work together

Phone a colleague

Write the code

Circle Image Chunking: Four Items

Write a script

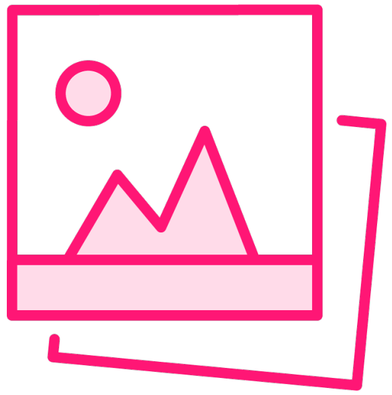
Review it

Record audio

Create a demo

Description Text and Three Item Icons with Headers

Add a short bit of description text in this area. Keep it short and simple.

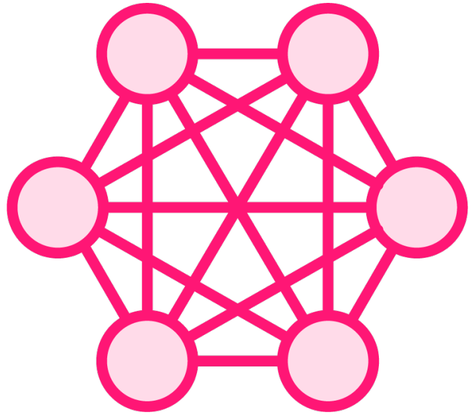


Analysis

Add a description here on these lines. Keep it short and simple.

Process

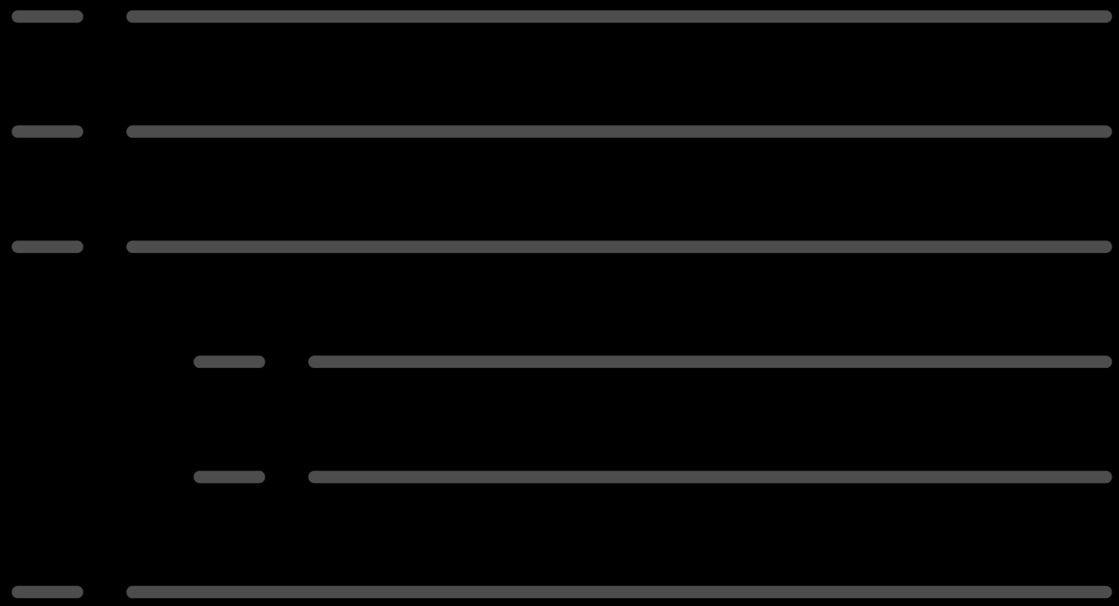
Add a description here on these lines. Keep it short and simple.



Work List

Add a description here on these lines. Keep it short and simple.

Bullet Lists



Animation built in

Bullet alternative

Sentence fragments

List of things

Procedure list

Talking points

Title or Relevant Graphic

Animation built in

Bullet alternative

Sentence fragments

List of things

Procedure list

Talking points

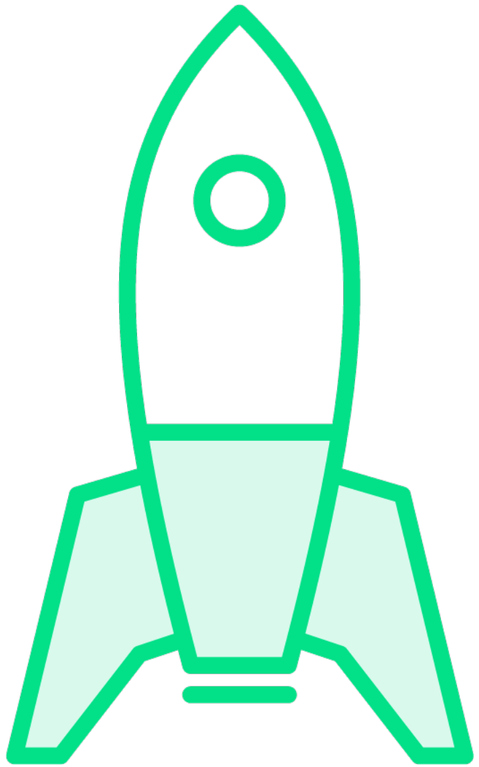
Title or Relevant Graphic

Animation built in

Room for a bit more text

Use this layout for:

- Longer sentence fragments
- List of things
- Procedure list
- Talking points



Animation built in

Room for a bit more text

Use this layout for:

- Longer sentence fragments
- List of things
- Procedure list
- Talking points

Title | Image Left | Bullets Right

Animation built in

Room for a bit more text

Use this layout for

- Longer sentence fragments
- List of things
- Procedure list
- Talking points

Title | Image Right | Bullets Left

Animation built in

Room for a bit more text

Use this layout for

Longer sentence fragments

List of things

Procedure list

Talking points

Photo Circle with Text

Works better with photos than with icons

Photos must be high-quality and copyright-safe

Re-center text boxes vertically once text is entered

The main image bleeds off three edges

Graphic should fill the entire space

- Graphic must be high-quality and copyright-safe

Text animation is built in

Left Image with Title

The main image bleeds off three edges

Graphic should fill the entire space

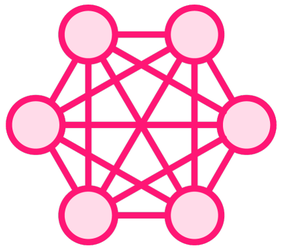
- Graphic must be high-quality and copyright-safe

Text animation is built in

Three Item List with Icons

An image chunking option for when you have longer text

Be concise and keep each bullet to 3 lines or fewer



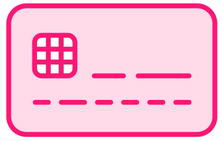
Add relevant icons on the left for each text item on the right

Four Item List with Icons

An image chunking option for when you have longer text

Be concise and keep each bullet to 2 lines or fewer

Add relevant icons on the left for each text item on the right



An alternative to bullets

Five Item List with Icons

An image chunking option for when you have longer text

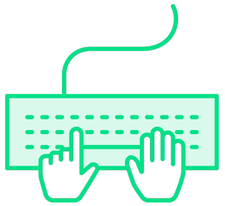
Sometimes four of these items aren’t enough

Add relevant icons on the left for each text item on the right

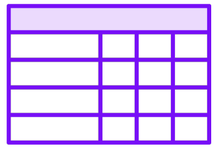
An image chunking option for when you have longer text

Be concise and limit the text to one line

Two Column Six Item List with Icons



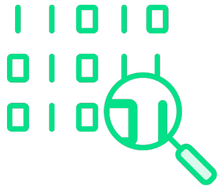
Program



Export all that data

Take a break and read

Write up a case study

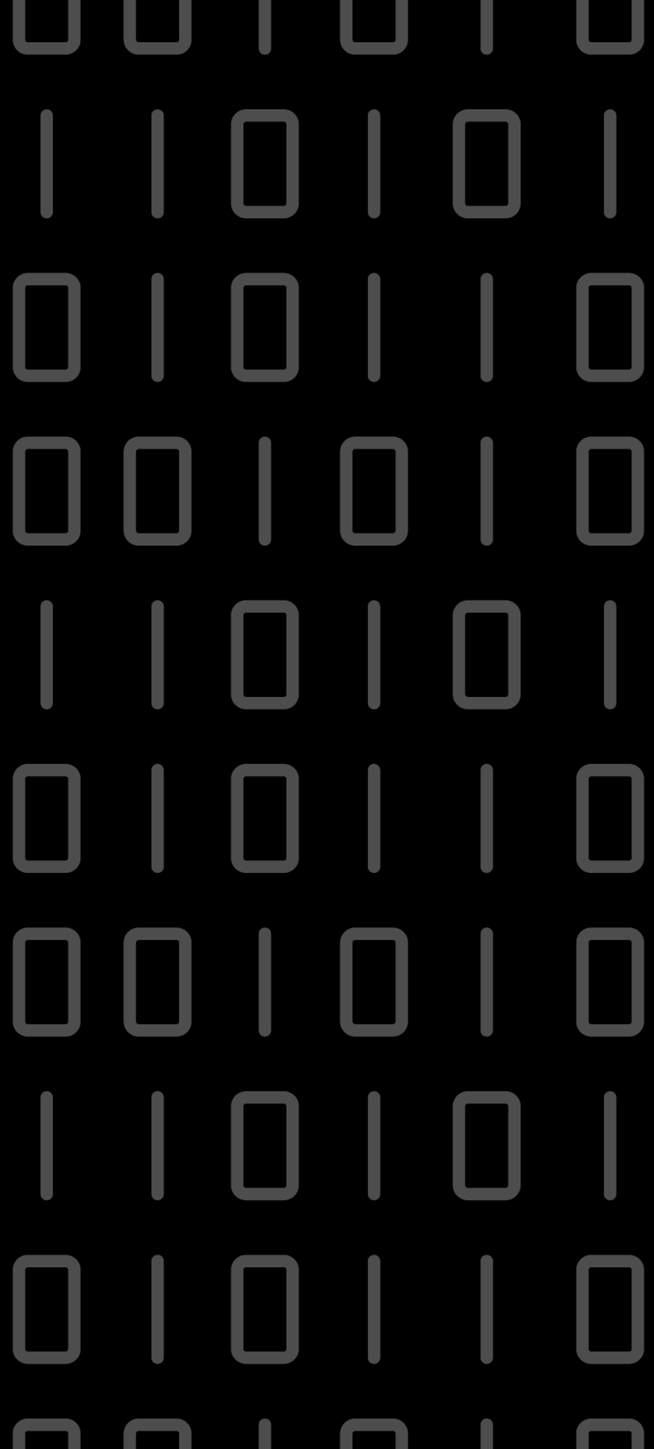


Check the data

It’s all under control

Code

Refer to Author Kit for additional guidance



Code Snippet | Light

```
<div class="row carousel-indicators">
    <div style="background-color:red;" class="col-md-4" data-
target="#homeCarousel" data-slide-to="0" class="active">
    </div>
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</div>
```

Code Snippet | Dark

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Code Snippet with Info | Light

Information about the code above. Code should be in Roboto Mono font, while slide title and supplemental text should remain in PS TT Commons.

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<div class="row carousel-indicators">
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</div>
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Code Snippet with Info | Dark

Information about the code above. Code should be in Roboto Mono font, while slide title and supplemental text should remain in PS TT Commons.

```
// Put code on this side

let website = {
  name: 'Pluralsight',
  url: 'https://pluralsight.com'
};

function makeUrl() {}

let sites = []
sites.push(website)

console.log(sites[0].name
```

- Line up with these notes
- Create an object
- Define a function
- Add the object to an array
- Log the site name to the console

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Title and Tab for Filename

Add a line of text here if needed

Filename.code

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```

Title and Two Tabs

Filename.here

```
<div style="background-color:red;"
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Title and Tab with Output | Vertical

Add a line of text here if needed

Filename.here

```
<div class="row carousel-indicators">
    <div style="background-color:red;" class="col-
md-4" data-target="#homeCarousel" data-slide-to="0"
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slide-to="1" class="active">
    </div>
</div>
```

```
> Console
```

Title and Tab with Output | Vertical

Filename.nere

```
<div class="row carousel-indicators">
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    </div>
</div>
```

```
> Console
```

Title and Tab with Output | Horizontal

Filename.here

```
<div style="background-color:red;"
  class="col-md-4" data-
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  class="active">
</div>
```

```
> Code output or other items
```

Title and Tab with Output | Horizontal

Filename.here

```
<div style="background-color:red;"
  class="col-md-4" data-
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Title Left and Filename at top

Filename.here

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<div class="row carousel-indicators">
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```

Title Left and Filename at top

Filename.here

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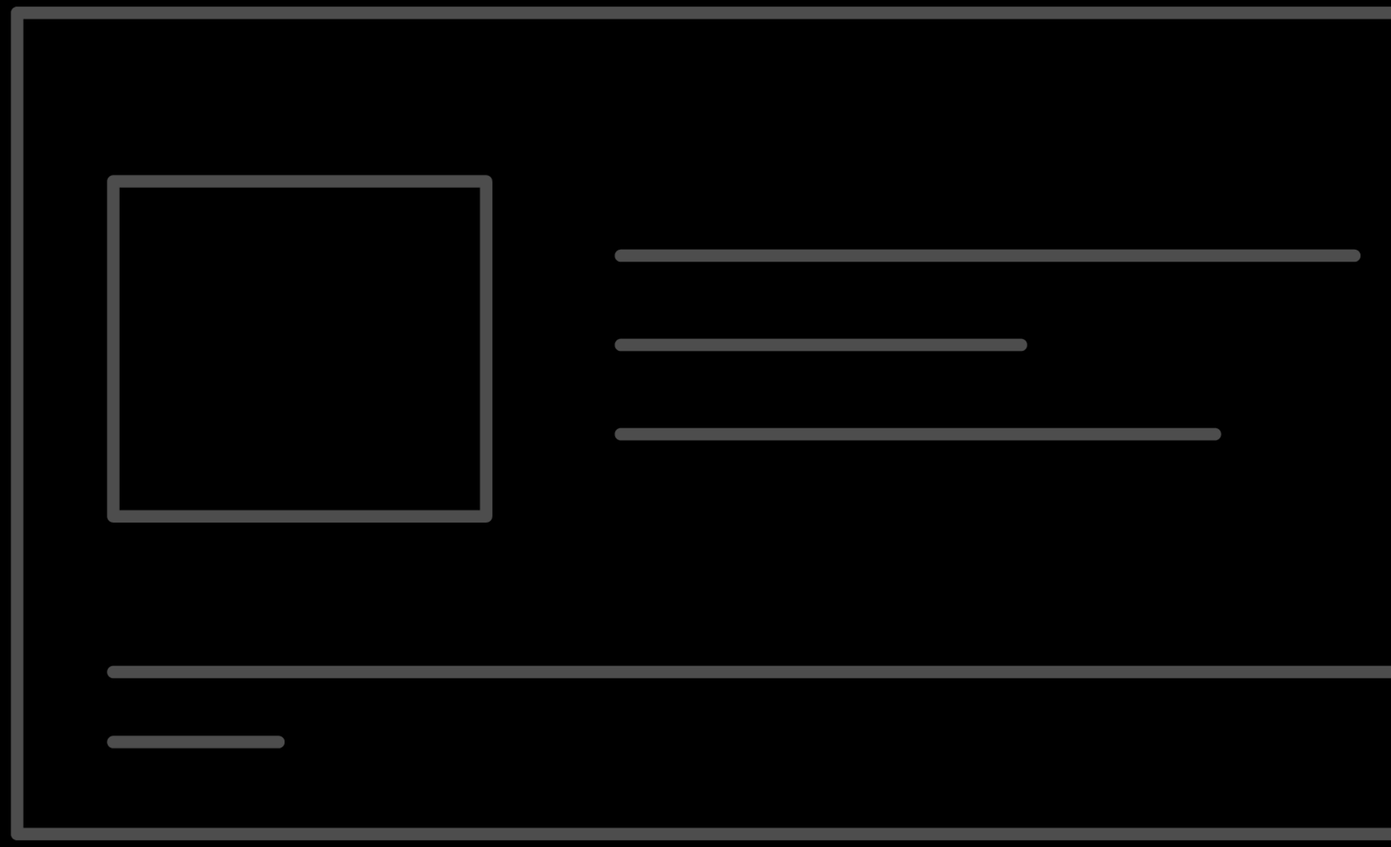
```

Citations

Refer to Author Kit for additional guidance



Citation Option

Citation: Author/Source, Title, Link/Short URL

More Information

Title of Pluralsight Course

Author Name

Resources Referenced in this Course

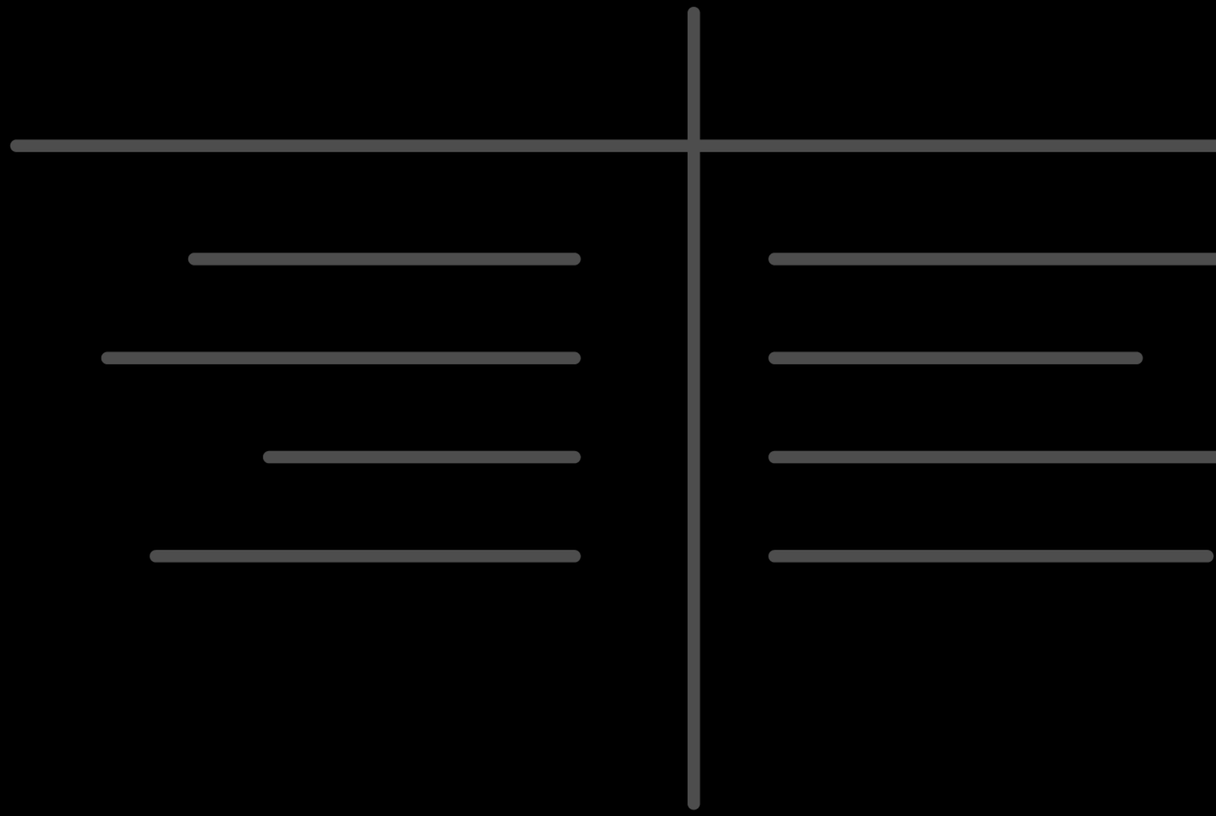
Microsoft Docs, Update Management Overview

Stephen Covey, The 7 Habits of Highly Effective People

Add relevant icons on the left for each text item on the right

Author/Source, Title, Link/Short URL

Comparison Slides



Comparison: Point-by-Point

Functional group	VS	Objectives
Configure and administer security		Manage vSphere storage
Configure advanced networking		Configure software-defined storage
Configure advanced storage		Configure vSphere storage multi-pathing and failover
Administer and manage resources		Perform advanced VMFS and NFS configurations and upgrades
Configure availability solution		Configure software-defined storage

Comparison: By Item

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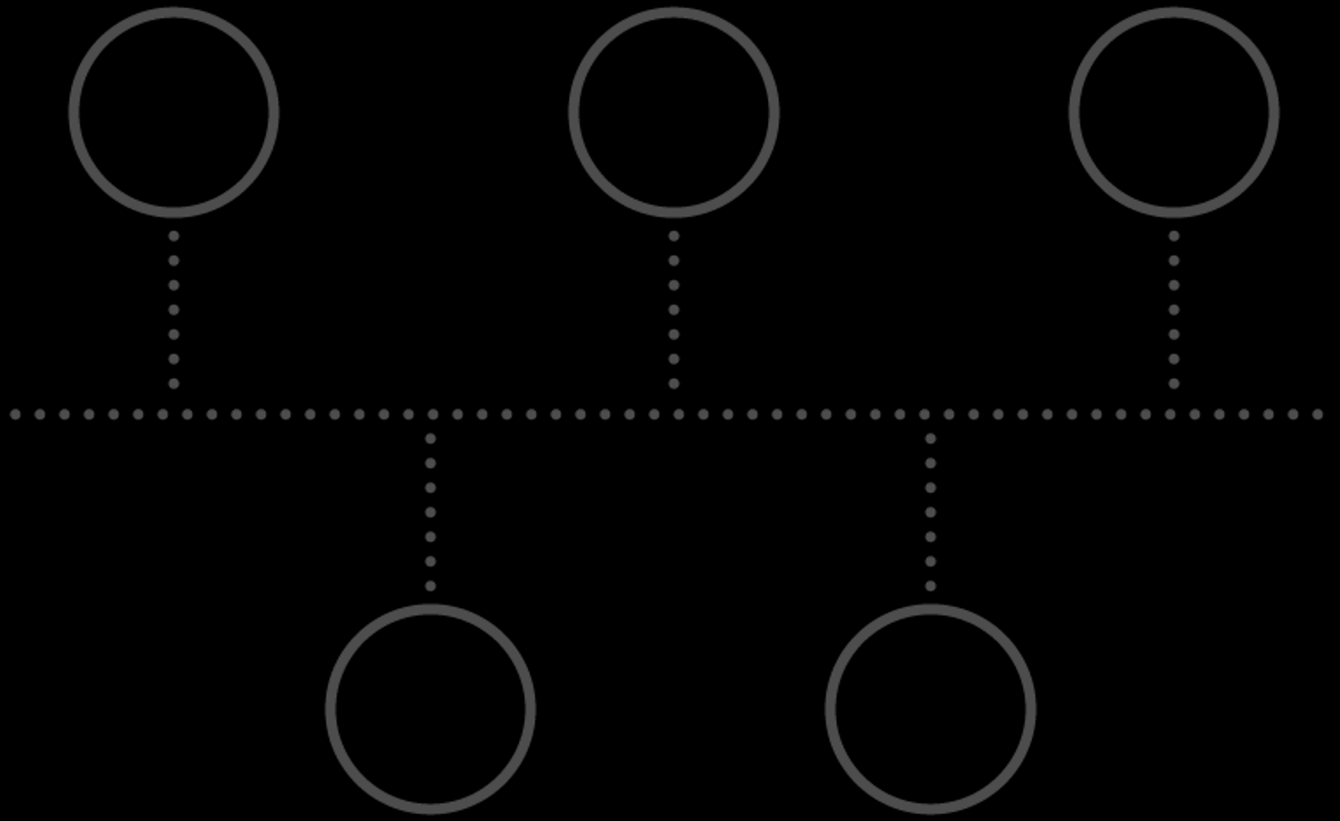
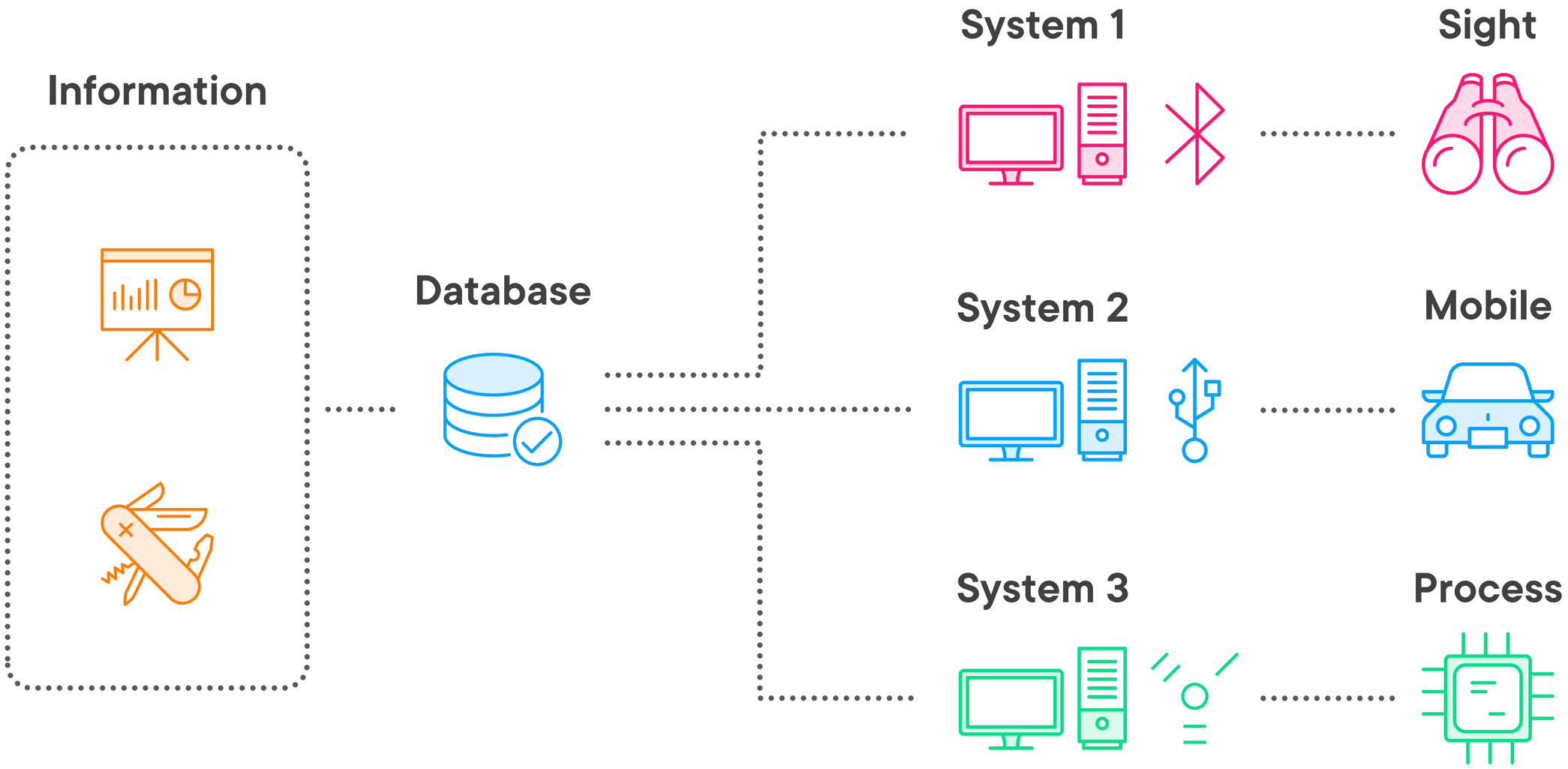
Configure availability solution		Configure software-defined storage

Diagrams

Refer to Author Kit for additional guidance



Basic Diagram



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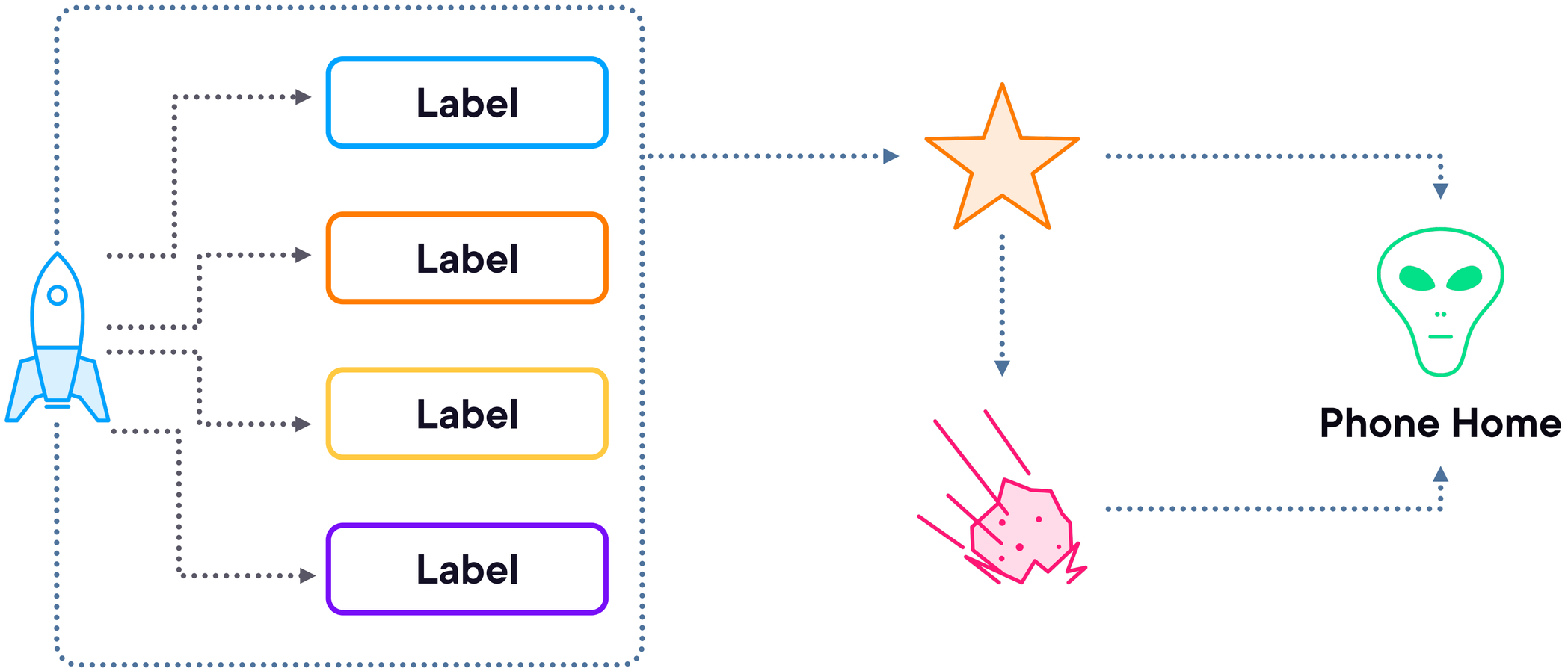

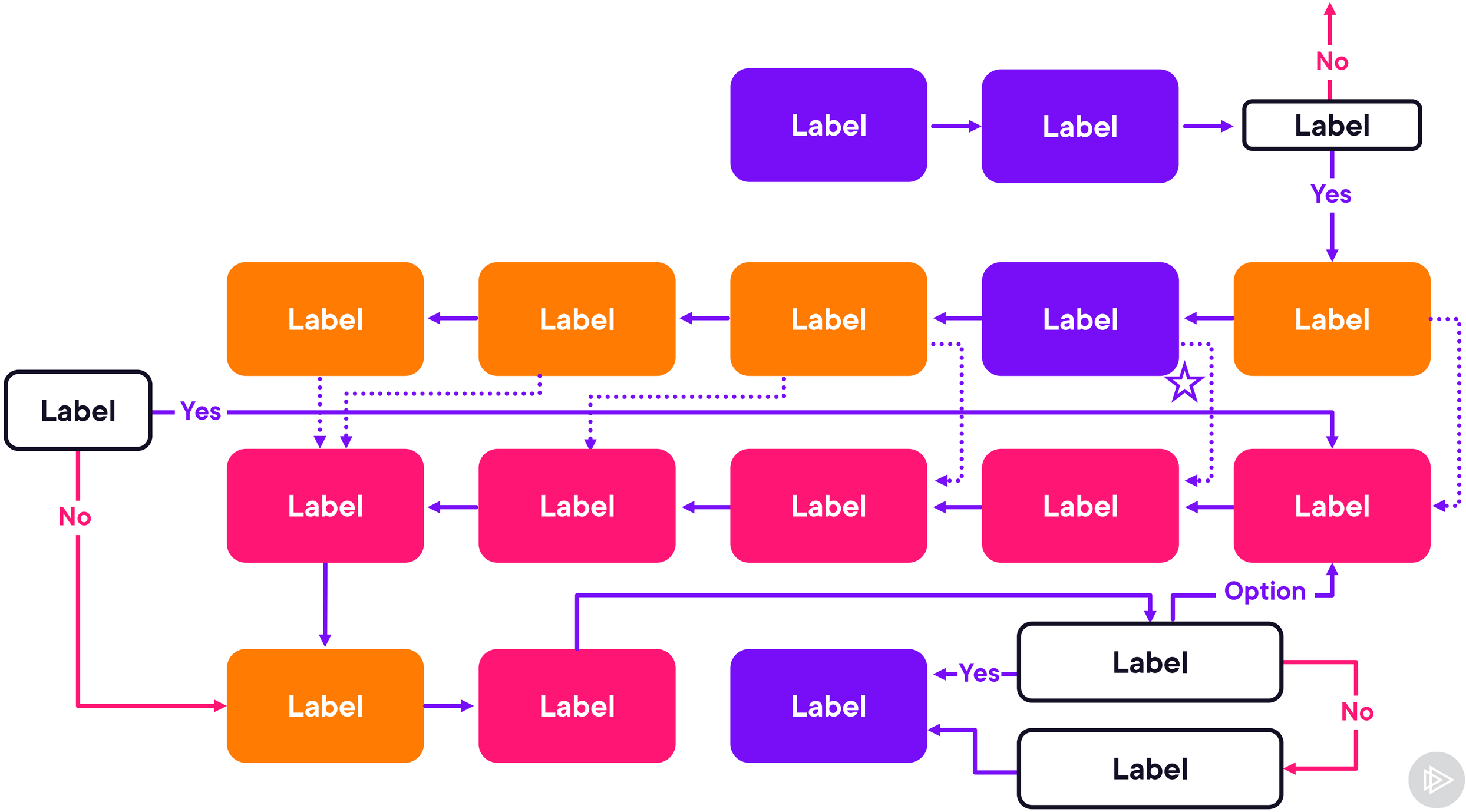
Label
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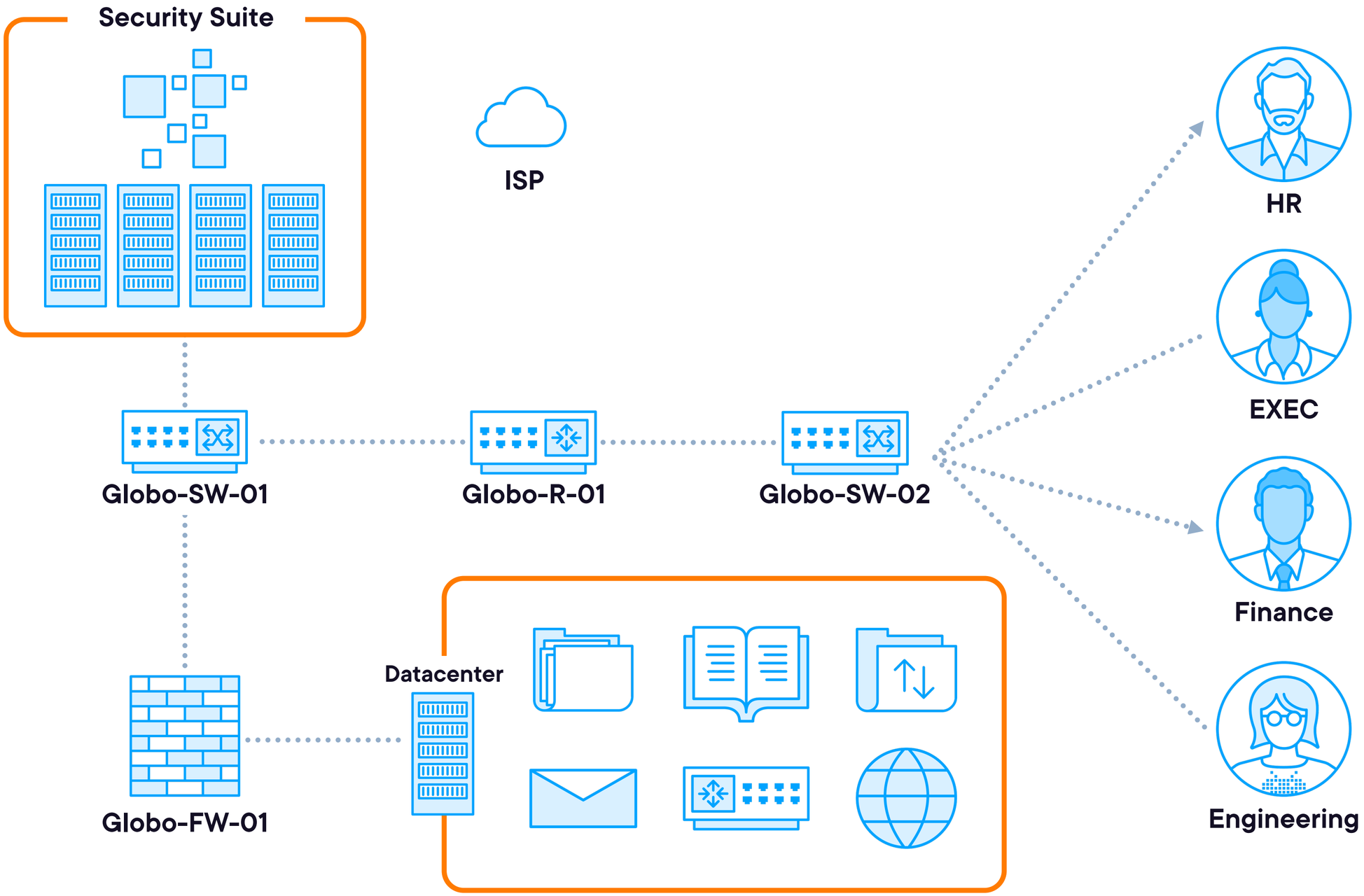
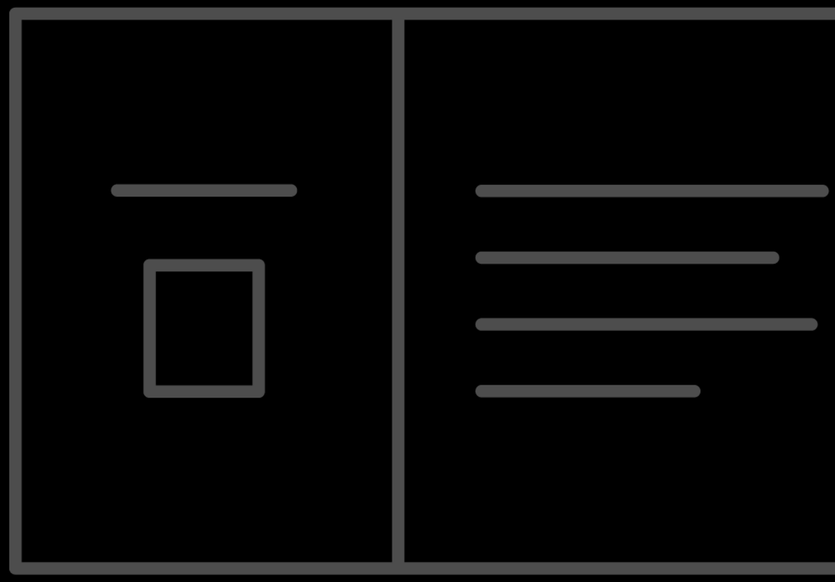
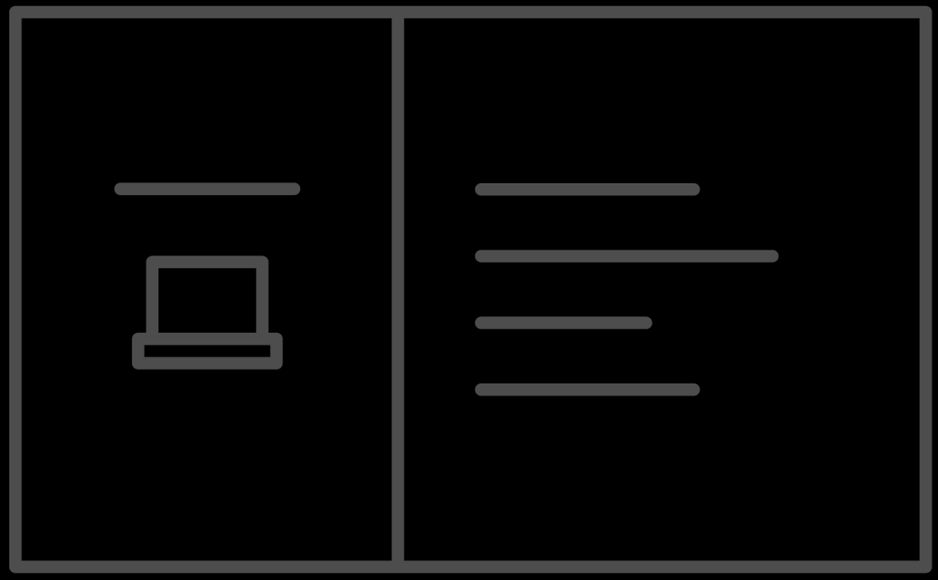

Security Suite
ISP
HR
EXEC
Finance
Engineering
Globo-SW-01
Globo-R-01
Globo-SW-02
Globo-FW-01
Datacenter

Slide Transitions

Refer to Author Kit for additional guidance





Timeline of Events with Fade Transition

1940s
McCulloch and Pitts
Neural network theory

1956
Stephen Cole Kleene
Regular events/sets

1968
Ken Thompson
First in computing

1973
Ken Thompson
First release of *grep*

1986
Henry Spencer
Regex library

1987
Larry Wall
Integration into Perl

*View in presentation mode

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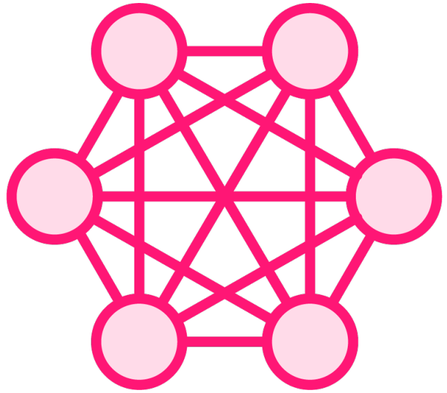
Cover/Uncover Transition

Some information about this graphic goes here

Some information about this graphic goes here



Some information about this graphic goes here

Make a statement or highlight an idea here between the transition.

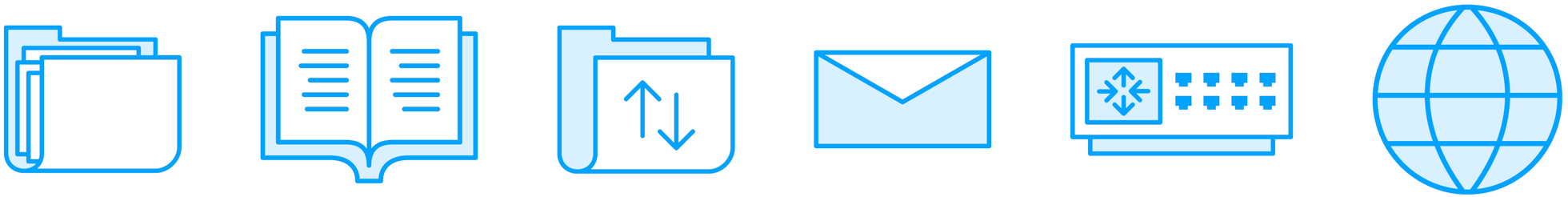
Push Transition: Step 1

Push transition: Let’s have lots of fun.

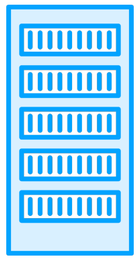
Push Transition: Step 2

There's so much you can do.

Morph Slide Transition



Datacenter

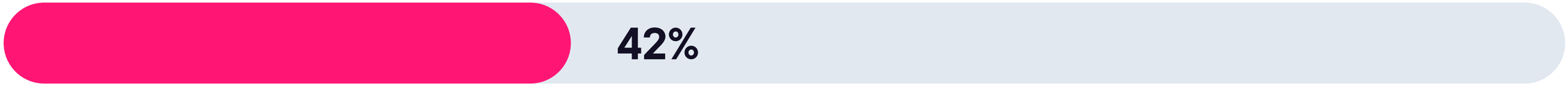


Morph Slide Transition

Morph Slide Transition

Q1

People who love pineapple on pizza



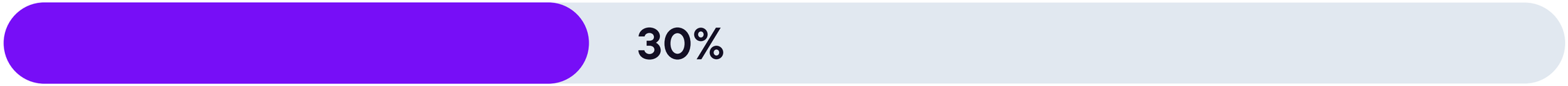
People who loathe pineapple on pizza

Morph Slide Transition

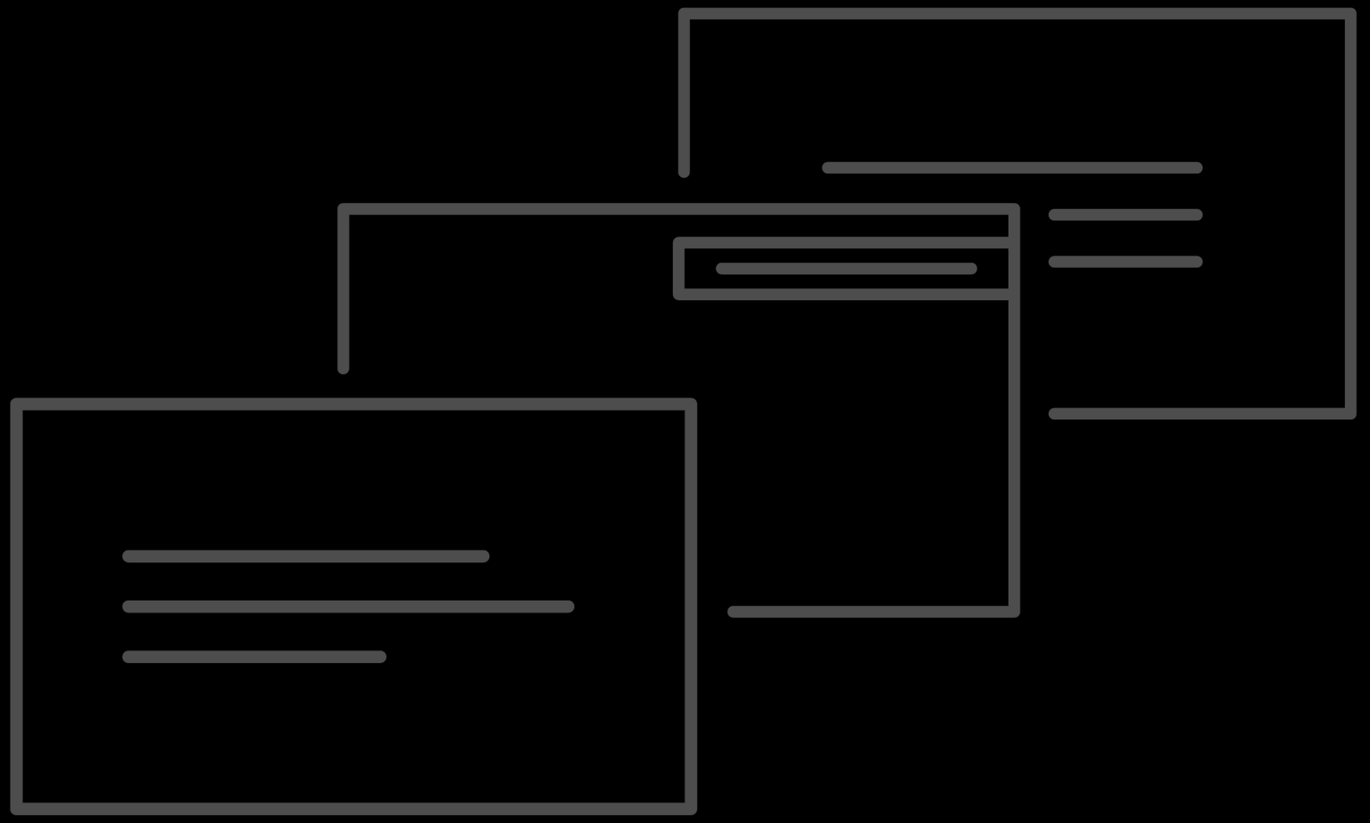
Q2

People who love pineapple on pizza

People who loathe pineapple on pizza



Other Slides



Word Definition

Here is where you put the definition. This is one of the few places where complete sentences are appropriate. Be sure to cite your source.

Citation: Author/Source, Title, Link/Short Url

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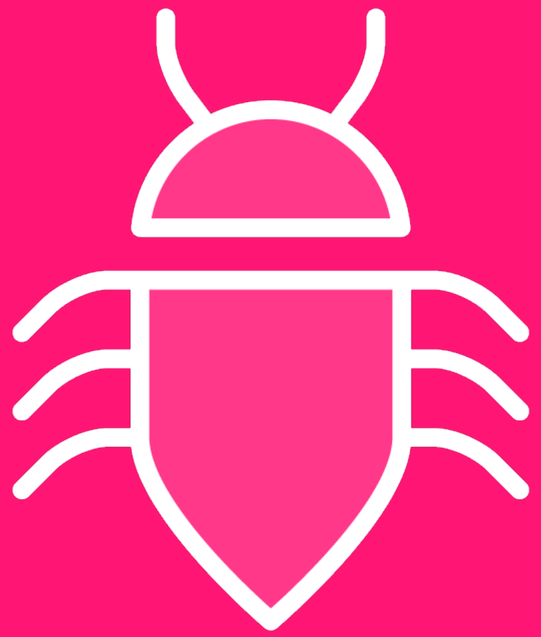
James Baldwin

“It is true that the more one learns the less one knows.”

Anaïs Nin

“Each friend represents a world in us, a world possibly not born until they arrive, and it is only by this meeting that a new world is born.”



Bugs, Bugs, Everywhere!

Add a short bit of description text in this area. This should be a quick intro slide to an idea without a lot of text. Stick to a few sentences.

Bugs, Bugs, Everywhere!

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This should be a quick intro slide to an idea
without a lot of text. Stick to a few sentences.

This is a short, important statement to bring attention to something.

You can use full-screen images too